



Magma Labs

STEM Opportunities

By Alex Li



Olympiads

Olympiads

- Math
 - AMC
 - AIME
 - USA(J)MO
 - MATHCOUNTS
- Programming
 - USACO
 - USAAIO
- Physics
 - F=MA
 - USAPHO



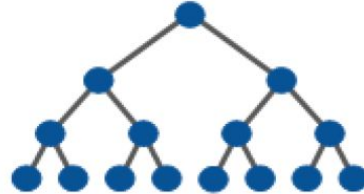
MAA AMC

American Mathematics Competitions



MATHCOUNTS[®]
COMPETITION SERIES

EST. 1983



USACO



Olympiads

Pros

- Lowest barrier to entry
 - Anyone can register for base olympiads
- Universal & standardized
 - 6 on AIME means the same everywhere
 - “Won a hackathon”
 - School hackathon (who cares?)
 - Company hackathon with 400 participants (impressive)
- Most prestigious at high levels
 - USA(J)MO = Top 300 in US
 - USACO Plat = Top 1300 in US

Competition	AIME Threshold	Honor Roll Distinction	Distinction
AMC 10 A	94.5	132	105
AMC 12 A	76.5	133.5	108
AMC 10 B	105	138	114
AMC 12 B	88.5	139.5	118.5



Olympiads

Why I quit olympiads

- After a certain point, olympiads become “grind problems”
 - Math: After AIME
- Olympiads are theoretical content
 - Comp math $\neq$ school math $\neq$ robotics math
 - CP $\neq$ Gamedev code || Webdev code





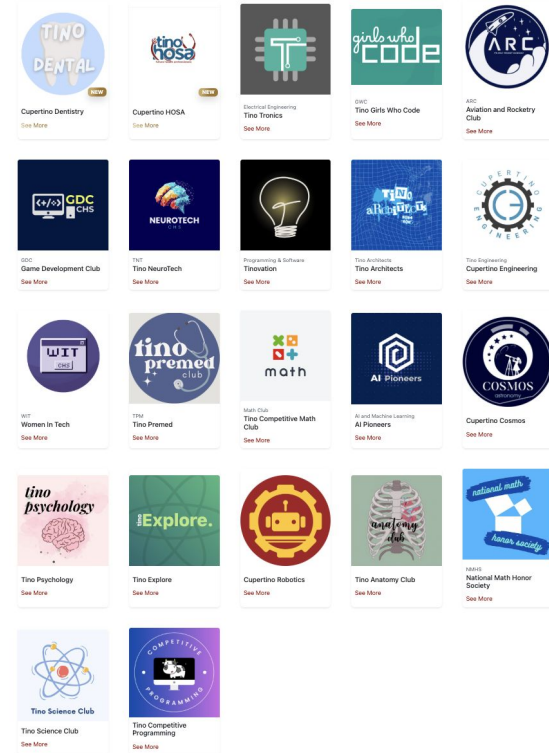
School Clubs

STEM clubs

School STEM clubs

- Coding
 - CS
 - Game/App/Web dev
 - CP
- Math
 - NMHS
 - Competitive math
- Other
 - Scibowl
 - Science Olympiad
 - Robotics

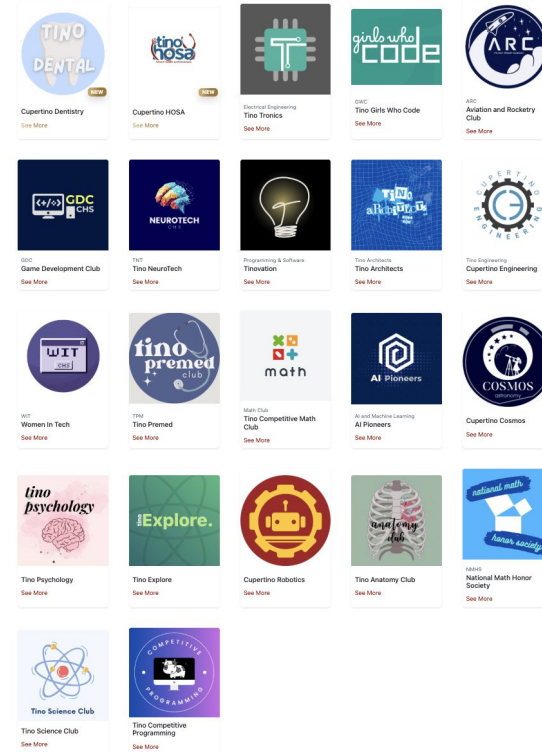
STEM & Technology Clubs



STEM clubs

- Officer positions are low barrier to entry
 - Most clubs are struggling to survive
 - All committed members become officers
- Relatively low time commitment
 - 3-6 hours a week
- Stable progression
 - Majority of officers become execs in their final year

STEM & Technology Clubs





Hackathons

Hackathons

- “Code a project in 12/24 hours”
- In person or online
 - In person events have free food
- School or company hosted
- Tracks
 - Educational Technology
 - Health and wellness
 - Social good



Hackathons

- Least time commitment
 - Hackathon season is spring
 - Hackathons are usually one full weekend day
- VERY heavily web/app dev focused
 - Nowadays very AI focused as well
- Wide range of events
 - School hackathons
 - Company hackathons
 - Hack Club
 - Flagship events
 - Satellite events





Robotics

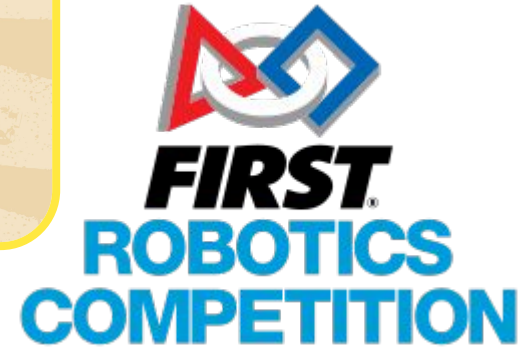
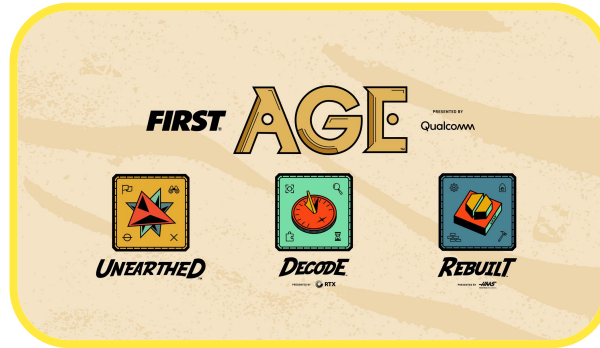
Robotics

- FIRST

- FRC
 - Largest robots
- FTC
 - More CV based
 - Smaller robots
- FLL (dying)
 - LEGO built robots

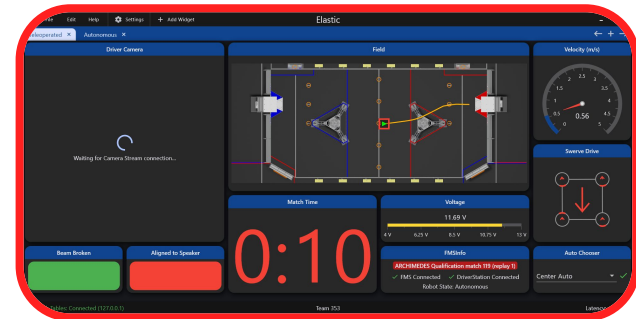
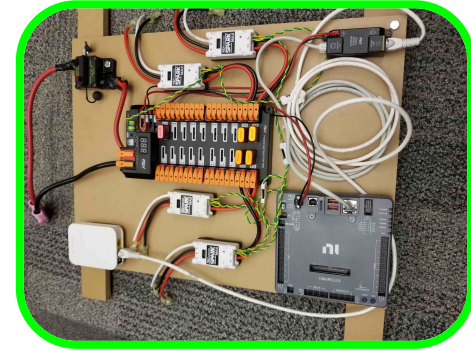
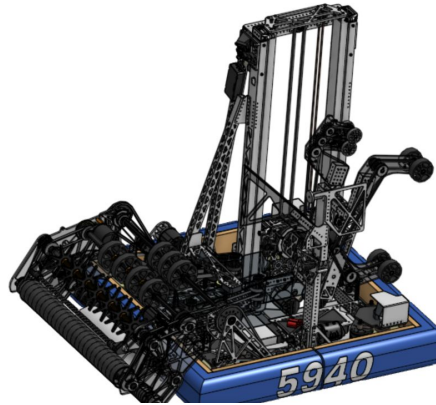
- VEX

- VEX IQ
- VEX V5
 - No custom parts allowed
 - FTC but cheaper and worse



Robotics

- Wide variety of roles
 - Hardware
 - Design
 - Fabrication
 - Electrical
 - Software
 - Auto/Teleop
 - Scouting
 - Business/Media
 - (for FRC) Impact Award
- School Teams
- Community Teams
- <https://www.firstinspires.org/team-event-search>



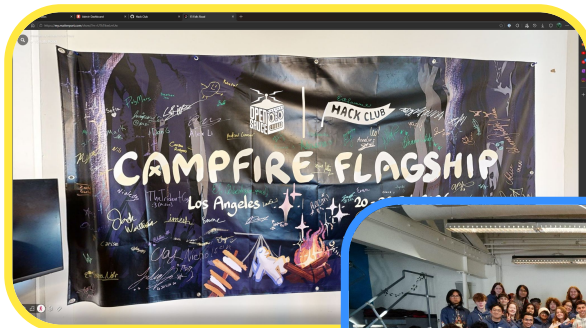


The Double Edged Sword of Hack Club

The Double Edged Sword of Hack Club

What is Hack Club?

- Nonprofit for teenagers by teenagers
 - “You Ship We Ship” programs
 - Build a project
 - Get funding/rewards
 - Go to flagship hackathons
 - Flagship Hackathons
 - 2-7 day long events
 - Well organized, themed, activities
 - Qualification requirement
 - Typically 60 hours of coding



The Double Edged Sword of Hack Club

What is Hack Club?

- Satellite Hackathons
 - Hackathon in 100+ cities across the globe on same weekend
 - Typically a game jam
 - One hack clubber in each city is chosen as the lead organizer
 - Lead org recruits HCCers as organizers
 - 2-3 month timeframe to organize event



The Double Edged Sword of Hack Club



Hacker Community

- Anything not open source is automatically the worst thing to ever exist
 - Apple MacBooks
 - Bambu Lab 3D printers
 - Unity Engine
 - Any OS other than some flavor of linux
- Assigning moral goodness to unnecessary work
 - Especially visible in hardware
 - “Why did you buy X when you can build it yourself?”



The Double Edged Sword of Hack Club

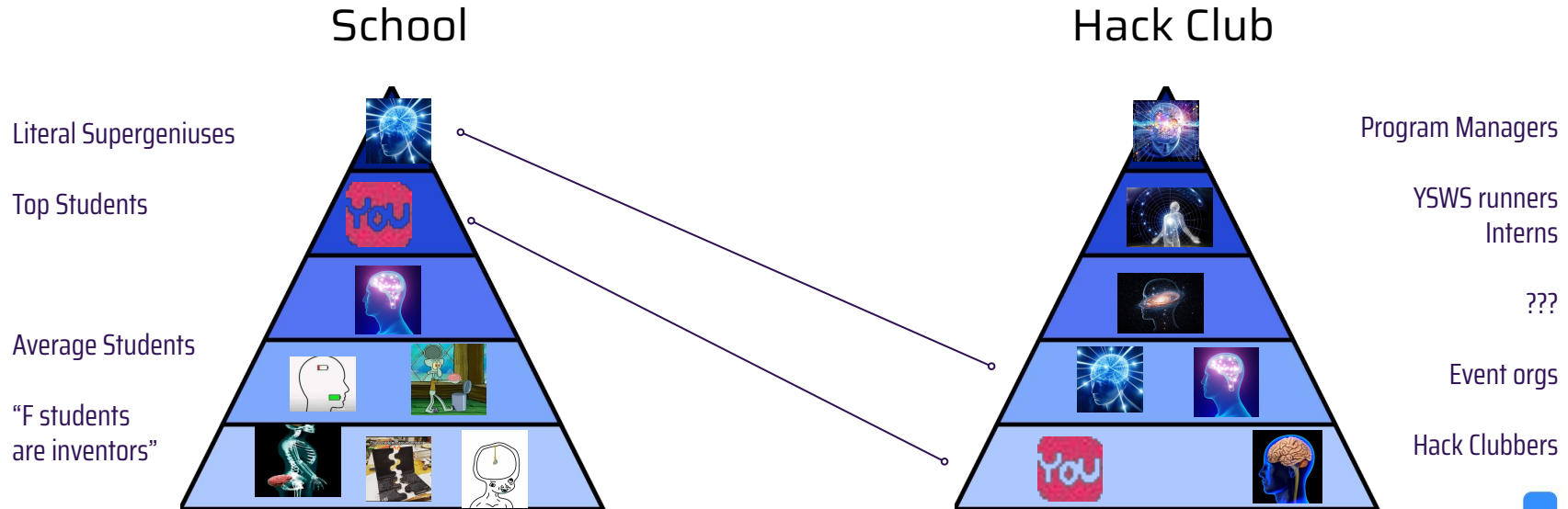


Hacker Community

- There are teenagers who
 - Build their own 3D printers
 - Organize flagship events
 - Start nonprofits
 - Speak onstage at CES
- You WILL feel stupid
 - Top student = dumb hack clubber
 - Valedictorian = average hack clubber
 - Top hack clubber = your Asian cousin



The Double Edged Sword of Hack Club



Hierarchy Pyramid - Higher is better





Conclusions

For college apps

	Barrier to entry	Time commitment	Prestige	Progression
Olympiads				
School Clubs				
Hackathons				
Robotics				

Comparison table - More spokes is better



General Tips

- You are a human
 - Don't do something just because it's prestigious
 - You WILL get burnt out
 - Enjoy what you do
- These are not your only choices
 - Other options
 - Non STEM options
 - Sports
 - Speech & Debate
 - FBLA
- Personal projects are amazing





Q&A